

Project Title:

Interactive Penguin Data Analysis Dashboard using Python Dash

Tools / Libraries Used:

- Python
- Dash
- Plotly Express
- Seaborn
- Pandas
- HTML Components (Dash)

Project Description :

This project presents an **interactive executive dashboard for penguin data analysis** built using **Python Dash and Plotly**. The dashboard analyzes the *Palmer Penguins dataset* and provides insights into penguin species, body mass, flipper length, and bill measurements. Users can dynamically filter the data by **species, island, and gender**, allowing real-time exploration of biological characteristics. The dashboard displays key performance indicators such as **total penguin count, average body mass, flipper length, and bill length**, along with visualizations including **species distribution, species comparison, relationship scatter plots, and correlation heatmaps**. This interactive system helps demonstrate **data visualization, exploratory data analysis, and dashboard development using Python**.

Code Implementation:

```
import dash

from dash import dcc, html, Input, Output

import plotly.express as px

import seaborn as sns

# Load & clean data

df = sns.load_dataset("penguins").dropna()

app = dash.Dash(__name__)
```

```

app.layout = html.Div([

    html.H1("Penguin Executive Dashboard", style={'textAlign':'center'}),

    # FILTERS
    html.Div([
        dcc.Dropdown(df['species'].unique(), multi=True, placeholder="Species", id='species_filter'),
        dcc.Dropdown(df['island'].unique(), multi=True, placeholder="Island", id='island_filter'),
        dcc.Dropdown(df['sex'].unique(), multi=True, placeholder="Gender", id='gender_filter')
    ], style={'display':'flex','gap':'15px','margin':'10px'}),

    # KPI CARDS
    html.Div([
        html.Div(id='total_penguins'),
        html.Div(id='avg_body_mass'),
        html.Div(id='avg_flipper'),
        html.Div(id='avg_bill'),
        html.Div(id='gender_split')
    ], style={'display':'flex','justifyContent':'space-around','margin':'20px'}),

    # CHART GRID
    html.Div([
        html.Div(dcc.Graph(id='distribution_chart'), style={'width':'48%'}),
        html.Div(dcc.Graph(id='comparison_chart'), style={'width':'48%'})
    ], style={'display':'flex','justifyContent':'space-between'}),

    html.Div([
        html.Div(dcc.Graph(id='relationship_chart'), style={'width':'48%'}),
        html.Div(dcc.Graph(id='correlation_chart'), style={'width':'48%'})
    ], style={'display':'flex','justifyContent':'space-between'})

```

```
])
```

```
@app.callback(
```

```
[
```

```
Output('total_penguins','children'),
```

```
Output('avg_body_mass','children'),
```

```
Output('avg_flipper','children'),
```

```
Output('avg_bill','children'),
```

```
Output('gender_split','children'),
```

```
Output('distribution_chart','figure'),
```

```
Output('comparison_chart','figure'),
```

```
Output('relationship_chart','figure'),
```

```
Output('correlation_chart','figure')
```

```
],
```

```
[
```

```
Input('species_filter','value'),
```

```
Input('island_filter','value'),
```

```
Input('gender_filter','value')
```

```
]
```

```
)
```

```
def update_dashboard(species, island, gender):
```

```
    dff = df.copy()
```

```
    if species:
```

```
        dff = dff[dff['species'].isin(species)]
```

```
    if island:
```

```
        dff = dff[dff['island'].isin(island)]
```

```
    if gender:
```

```
        dff = dff[dff['sex'].isin(gender)]
```

```

total = f"Total Penguins: {len(dff)}"

avg_mass = f"Avg Body Mass: {round(dff['body_mass_g'].mean(),2)}"

avg_flipper = f"Avg Flipper Length: {round(dff['flipper_length_mm'].mean(),2)}"

avg_bill = f"Avg Bill Length: {round(dff['bill_length_mm'].mean(),2)}"


gender_pct = dff['sex'].value_counts(normalize=True)*100

gender_split = f"Male %: {round(gender_pct.get('Male',0),1)} | Female %:
{round(gender_pct.get('Female',0),1)}"


dist = px.bar(dff, x='species', title="Species Distribution")


comp = px.bar(
    dff.groupby('species')['body_mass_g'].mean().reset_index(),
    x='species', y='body_mass_g',
    title="Avg Body Mass by Species"
)


rel = px.scatter(
    dff, x='flipper_length_mm', y='body_mass_g',
    color='species',
    title="Flipper vs Body Mass"
)


corr = px.imshow(
    dff.corr(numeric_only=True),
    text_auto=True,
    title="Correlation Heatmap"
)


return total, avg_mass, avg_flipper, avg_bill, gender_split, dist, comp, rel, corr

```

```
if __name__ == '__main__':
    app.run(debug=True)
```

Output:



Penguin Executive Dashboard



Penguin Executive Dashboard

